



Ze Zhuang Liu

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Research Interests: Industrial Ecology, Urban Metabolism, Dynamic Material Flow Analysis, Spatial Life Cycle Assessment

Professional Profile: Interdisciplinary researcher specializing in regionalized urban metabolism and prospective environmental assessment. Demonstrated project leadership and independent capability in delivering high-impact scientific outputs (e.g. PNAS, ES&T).

Proficient in integrating advanced geospatial intelligence (R, QGIS) with Machine Learning and dynamic MFA/LCA. Technical expertise encompasses modeling large-scale built-environment material stocks, climate-resource nexus, and circular economy strategies, directly supporting sustainable built-environment and disaster waste management.



重庆大学
CHONGQING UNIVERSITY

Education

Chongqing University, College of Environment and Ecology (Sep. 2024 – Jun. 2027, Expected)

M.S. in Environmental Science and Engineering; Weighted Score: 89/100; Advisor: Prof. Xiaofeng Gao

Chongqing University, College of Environment and Ecology (Sep. 2020 – Jun. 2024)

B.E. in Environmental Engineering; Weighted Score: 87/100 (Top 4%)

Publications & Projects

1. Jian Sun, **Ze Zhuang Liu (Co-first author, 1st among students)**, et al. Uncovering the Impacts of 2D and 3D Urbanization on Urban Heat Islands in 384 Chinese Cities. (*Environ. Sci. Technol.* 2025, 59, 14, 7106–7116).
2. Jian Sun, **Ze Zhuang Liu (2nd, 1st among students)**, et al. Uncovering the Nexus between Urban Heat Islands and Material Stocks of Built Environment in 335 Chinese Cities. (*Environ. Sci. Technol.* 2024, 58, 31, 13760–13771).
3. Jian Sun, ..., **Ze Zhuang Liu (3rd author)**, et al. Gridded material stocks in China based on geographical and geometric configurations of the built-environment. (*Scientific Data*, 2023, 10, 915).
4. **Ze Zhuang Liu (First author)**, et al. Global accumulation and persistent inequality of material stocks from 1970 to 2100. (Target journal: *the Proceedings of the National Academy of Sciences*, To be submitted).
5. **Ze Zhuang Liu (First author)**, et al. Topographic constraints on regional urban metabolism and material stocks in the Sichuan-Chongqing Region. (Target journal: *Resources, Conservation & Recycling*, To be submitted)
6. Jian Sun, **Ze Zhuang Liu (2nd, 1st among students)**, et al. Building and human safety: earthquake impacts and their mitigation in China. (*the Proceedings of the National Academy of Sciences*, Under review).
7. Predicting Earthquake Disaster Waste and Low-Carbon Management Strategies in the Sichuan-Chongqing Region (**Principal Investigator**, Chongqing Municipal Postgraduate Research and Innovation Project).

Core Technical Skills

1. **Geospatial Intelligence & Spatial LCA:** Mastery of R (terra, sf, tidyverse) and QGIS for PB-scale raster/vector processing; expert in pixel-level spatial analysis (Pub. 1, 2) and integrating spatial data with environmental assessments.
2. **Systems Analysis & Circular Economy:** Proficient in Top-down (Pub. 4) and Bottom-up (Pub. 3, 5) dynamic Material Flow Analysis (MFA); expert in LCA, NSGA-II multi-objective optimization, and construction & demolition waste (CDW) management strategies (Project 7).
3. **Data Mining & ML:** Expert in XGBoost and Random Forest (Pub. 1, 2, 4, 5); advanced in SHAP explainable attribution (Pub. 4, 5) and K-means clustering (Pub. 4). Proficient in Python scraping and Ollama-based LLM/NLP processing.
4. **Risk Assessment:** Mastery of probabilistic modeling for building vulnerability and large-scale population exposure risk assessment (Pub. 6, Project 7).

Honors & Leadership

Honors: National Scholarship (2021); Gold Award in the 8th China International “Internet+” Competition (2022); Second Prize in the National Mathematical Modeling Competition (2022).

Leadership: Executive President of the College Student Union (2022-2023), managed large-scale academic and social events for 1,000+ participants.